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Toward digital twin-enabled real-time structural damage detection via POD-assisted Bayesian estimation

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Abstract

Accurate and timely detection of structural damage is essential for ensuring the safety and integrity of civil infrastructure. While Bayesian filtering has been widely adopted in structural health monitoring (SHM) for estimating system states and parameters, full-scale parameter estimation can often result in biased outcomes and significant computational costs. To overcome these limitations, this study proposes a framework toward digital twin-enabled SHM by integrating proper orthogonal decomposition (POD) with Bayesian filtering. The framework uses displacement and excitation measurements from the system to estimate states in real-time, updated proper orthogonal mode (POM) slopes, and the location and severity of damage. Structural damage localization is achieved by continuously monitoring changes in the slope of the first POM, which is updated in real time using an online singular value decomposition (SVD) algorithm. Simultaneously, the Unscented Kalman Filter (UKF) estimates system states and selectively updates stiffness parameters, focusing only on degrees of freedom identified as damaged by comparing the evolving POM slopes with a baseline reference derived from the healthy structure. This coordinated operation of online SVD and UKF enables efficient, accurate, and robust assessment of structural condition in real time. The proposed framework addresses three critical stages of SHM that is, detection, localization, and quantification. Its performance is validated through comprehensive numerical simulations on an eight-story shear frame under various damage scenarios, including adjacent-story, non-adjacent, and progressive damage cases. Furthermore, validation on a laboratory-scale three-dimensional steel frame confirms its practical applicability. The results demonstrate that the proposed approach can reliably localize and quantify multiple damage events, offering a scalable solution for real-time SHM.

Keywords

Digital twin, system identification, damage detection, reduce order model, online proper orthogonal decomposition, Bayesian filtering

Introduction

A digital twin (DT) is a virtual replica of a physical system that uses real-time sensor data to reflect its behavior and evolving state. Advances in the Internet of Things (IoT), machine learning, and computational modeling have established DTs as transformative tools across engineering disciplines. Glaessgen and Stargel[1] define a DT as "an integrated multiphysics, multiscale, probabilistic simulation of an as-built system that uses the best available physical models, sensor updates, fleet history, etc., to mirror the life of its corresponding physical twin." A DT typically integrates three core components: physical system data, physical or surrogate models, and synchronization mechanisms to align the virtual and physical domains.[2,3] Applications span manufacturing,[4] healthcare,[5] smart cities,[6,7] environmental monitoring,[8] and civil infrastructure.[9–11] In civil infrastructure, DTs show significant promise for structural health monitoring (SHM), where integrated data-driven and physics-based methods enable continuous assessment of structural integrity.

Among various SHM approaches, vibration-based techniques are widely adopted in SHM due to their non-invasive nature and sensitivity to structural changes. These methods rely on dynamic

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characteristics such as natural frequencies, mode shapes, and damping ratios to detect damage.[12,13] Modal-based damage detection remains popular for its simplicity, interpretability, and effectiveness. While shifts in natural frequencies can signal damage by indicating stiffness loss, their sensitivity is limited for small or localized damage,[14] therefore, mode shapes and their spatial derivatives particularly mode shape slope and curvature are preferred for accurate localization. Early studies[15] demonstrated that curvature mode shapes exhibit strong sensitivity to local stiffness reduction and can effectively indicate damage location. Other studies have validated the effectiveness of curvature-based methods through extensive numerical and experimental investigations, further establishing their robustness for practical damage detection applications.[16] Later investigations[17] compared curvature-based indicators with damage locating vector methods and confirmed the strong localization capability of curvature features across different damage scenarios. Subsequent research has further refined curvature-based approaches through enhanced numerical and analytical formulations to improve damage localization accuracy under weak damage scenarios.[18] Reviews have emphasized the importance of such mode-shape-based indicators in capturing subtle integrity changes.[19–23] Their use spans various structures, including beams,[24–26] shear buildings,[27] laminated plates,[28] and general plate structures.[29] Techniques such as modal strain energy with hybrid optimization have enhanced damage detection in steel frames,[30] while mode shape slope and curvature-based formulations have enabled localization and severity estimation of damaged zones.[31,32] Perturbation-based and slope ratio approaches further highlight strong correlations between damage and changes in the fundamental mode shape.[27,32] Although higher modes offer increased sensitivity to localized damage,[33,34] their extraction in real-world conditions remains challenging.[35]

Extending the concept of mode shape-based damage detection, data-driven methods have been introduced that utilize similar spatial indicators such as slope and curvature but apply them to statistical vibration modes extracted directly from measured data. One such technique is proper orthogonal decomposition (POD), which yields a set of orthogonal spatial patterns known as proper orthogonal modes (POMs). POM derivatives (slope and curvature) effectively highlight localized damage through abrupt spatial variations. Similar to mode shape curvature in traditional analysis, POM curvature enhances sensitivity to damage. For instance, POD has been integrated with optimization for damage localization in shear type buildings[36] and with POMs to identify damage in composite beams,[37] treating POMs as dynamic invariants to reduce environmental influence. POM-based features have been implemented in a field-programmable gate array-enabled unsupervised SHM framework for a bridge structure, demonstrating their effectiveness in identifying structural damage.[38] A baseline-free method using POM changes in guided wavefields showed high accuracy across damage scenarios.[39] POMs from frequency response functions have also proven effective in beam damage detection.[40] Additionally, POD combined with neural networks has enabled damage detection in bridges and buildings, demonstrating strong adaptability across structural types and data conditions.[41–43]

Although modal and POD-based techniques effectively detect structural damage, their offline nature limits real-time SHM applications. To overcome this, Bayesian filtering methods offer recursive, real-time estimation of states and parameters.[9,44,45] The Kalman Filter (KF)[46] provides optimal estimation for linear systems with known inputs and sparse measurements and is widely applied in response reconstruction and state estimation.[47–49] However, its linearity assumption limits use in nonlinear systems. The Extended Kalman Filter (EKF) addresses this by linearizing dynamics around current estimates.[50] EKF has been used for fatigue life prediction in plates under coupled loading[51] and for modal identification in systems with tuned mass dampers,[52] though its performance degrades in strongly nonlinear cases. To improve robustness, the Unscented Kalman Filter (UKF) was developed,[53–55] employing the unscented transform (UT) to capture nonlinearities more accurately without linearization. UKF maintains recursive estimation and suits nonlinear SHM applications. Recent developments include DT frameworks using KFs for time-variant model identification via GSINDy,[56] validated on the Lorenz system and an industrial hydrotreating unit. Other DTs using KFs have been proposed for state monitoring and anomaly detection,[57] and for joint input-state-parameter estimation via nonlinear minimum variance unbiased filters validated with vision-based measurements.[58]

Hierarchical Bayesian filtering using augmented Kalman filters (AKF) and particle filters (PF) has been integrated with parametric reduced-order models to enable joint state, input, and parameter estimation under output-only measurements.[59] AKF has also been coupled with a substructuring approach to estimate strain responses at unmeasured locations of a rollercoaster connection for remaining fatigue life prediction.[60] Gaussian process latent force models have been employed for virtual sensing in wind turbine towers, providing stable and accurate estimation of dynamic strains and damage-equivalent loads compared with conventional KF formulations.[61] Despite their advantages, Bayesian filters face challenges as the number of

unknown parameters increases, often leading to biased and unstable estimates.[62] To address this, POD has been integrated with Bayesian filtering for dimensionality reduction. A combined POD–Bayesian filtering framework was proposed for real-time SHM in a reduced-order space,[63] where a KF updated the POMs and an EKF estimated states and parameters. Damage localization relied on visual changes in POMs. Similarly, a real-time SHM method using online singular value decomposition (SVD) and EKF was developed,[64] with POMs updated via SVD and EKF operating in the reduced space. While effective for tracking stiffness changes, these approaches estimate in a transformed domain, lacking direct physical interpretability. For practical SHM, especially in real-time diagnostics, translating estimates to the physical coordinate space remains critical.

Traditional Bayesian filtering techniques, though effective for real-time state and parameter estimation, are computationally intensive as they track all parameters continuously. Similarly, existing POD Bayesian frameworks rely on reduced-order models lacking physical interpretability. To overcome these limitations, this study presents a framework toward DT-enabled SHM by combining online POD with Bayesian filtering for real-time SHM. Unlike conventional POD approaches, the method uses online SVD to extract POMs and track their slopes relative to a healthy baseline as damage-sensitive features in online manner. Damage detection triggers localized Bayesian parameter estimation, reducing computational cost and retaining physical interpretability. This adaptive strategy enables efficient, real-time monitoring and accurate localization of structural damage, as demonstrated through numerical and experimental validation.

Problem statement

A linear dynamic system with N degree of freedom (DOF) is described by the following second-order differential equation:

$$\mathbf{M}(t)\ddot{\mathbf{x}}(t) + \mathbf{C}(t)\dot{\mathbf{x}}(t) + \mathbf{K}(t)\mathbf{x}(t) = \mathbf{F}(t) \tag{1}$$

In this equation, $\mathbf{M}(t), \mathbf{C}(t)$ and $\mathbf{K}(t) \in \mathbb{R}^{N\times N}$ represent the mass, damping, and stiffness matrices, respectively, are time-dependent to reflect the changes in the system properties caused by damage progression. The vectors $\mathbf{x}(t), \dot{\mathbf{x}}(t)$ and $\ddot{\mathbf{x}}(t) \in \mathbb{R}^{N\times 1}$ denote displacement, velocity, and acceleration, while $\mathbf{F}(t) \in \mathbb{R}^{N\times 1}$ is the external force vector.

This study assumes that structural damage is introduced through a reduction in stiffness, while the mass and damping matrices remain unchanged throughout the analysis. Estimating all stiffness parameters along with the full state vector is computationally intensive, and Bayesian filters often face challenges in maintaining accuracy when applied to high-dimensional parameter spaces.

To overcome these limitations, the proposed framework integrates online-POD with UKF in an online manner. POD enables real-time damage localization by analyzing structural response data, while UKF simultaneously estimates system states and updates only the stiffness parameters at the identified damage locations. This targeted strategy avoids full parameter estimation, reduces computational cost, and improves accuracy in real-time SHM.

POD-assisted Bayesian estimation

Proper orthogonal decomposition

POD is used to transform high-dimensional structural response data into a lower-dimensional form, capturing the most significant features of the system.[65–67] For structural systems, POD identifies dominant spatial patterns in the dynamic response, offering insight into how the structure behaves under varying conditions. Let the displacement responses from N DOF be represented as $\{\mathbf{x}_1(t), \mathbf{x}_2(t), \ldots, \mathbf{x}_N(t)\}$, collected and stored in the data matrix $\mathbf{D}$:

$$\mathbf{D} = \begin{bmatrix} x_1(t_1) & x_1(t_2) & \cdots & x_1(t_n) \\ x_2(t_1) & x_2(t_2) & \cdots & x_2(t_n) \\ \vdots & \vdots & \ddots & \vdots \\ x_N(t_1) & x_N(t_2) & \cdots & x_N(t_n) \end{bmatrix}_{N\times n} \tag{2}$$

where n is the number of snapshots, that is, the discrete time samples of the system's response recorded at successive time instants during a single dynamic simulation or experiment.

The covariance matrix $\mathcal{C}$ is computed as:

$$\mathcal{C} = \frac{1}{n}\mathbf{D}\mathbf{D}^T \tag{3}$$

By applying eigenvalue decomposition, the dominant spatial modes (POMs) can be obtained as:

$$\mathcal{C}\mathbf{\Phi} = \mathbf{\Phi}\mathbf{\Lambda} \tag{4}$$

In this study, SVD is used to perform POD, it can be given as:

$$\mathbf{D} = \mathbf{\Phi}\mathbf{\Lambda}\mathbf{V}^T \tag{5}$$

In this decomposition, $\mathbf{\Phi} = [\phi_1, \phi_2, \ldots, \phi_N] \in \mathbb{R}^{N\times N}$ is the left singular matrix also known as POMs, $\mathbf{V} \in \mathbb{R}^{n\times n}$ is right singular matrix and $\mathbf{\Lambda} \in \mathbb{R}^{N\times n}$ is a pseudo-

diagonal matrix of singular values. In practice, it is not necessary to retain all N modes. The number of retained modes $\eta \ll N$ depends on the cumulative energy they capture, calculated as:

$$\frac{\sum_{i=1}^{\eta} \Lambda_{ii}^2}{\sum_{i=1}^{N} \Lambda_{ii}^2} \geqslant \gamma \tag{6}$$

where $\gamma \in [0, 1]$ is the energy threshold; values of γ approaching one signify that the retained modes contain most of the system's dynamic features.

In an undamaged structure, the POM typically shows a smooth spatial pattern, reflecting the uniform dynamic properties of the system. When damage is present, this uniformity is disturbed. The change alters the energy distribution in the system and introduces localized irregularities in the POM, such as kinks or sharp gradients, near the damaged region.[63] While these irregularities are useful indicators, they may be subtle. To improve sensitivity, the slope of the POM is analyzed, as it magnifies these local deviations and better highlights potential damage locations.[31,68] The slope of a POM $\boldsymbol{\phi}_i$ is defined as its spatial derivative as $\frac{\partial \boldsymbol{\phi}_i}{\partial y}$.

For discrete data, this derivative can be approximated using the forward difference method:

$$\left.\frac{\partial \boldsymbol{\phi}_i}{\partial y}\right|_j \approx \frac{\phi_i(y_{j+1}) - \phi_i(y_j)}{y_{j+1} - y_j} \tag{7}$$

Damage in the structure results in sharper gradients or noticeable changes in the slope profile. These deviations are more pronounced than changes in the POM itself and serve as effective indicators for identifying damage locations. In this study, an online SVD approach is employed to compute the POM in real time. This enables continuous damage localization as the system evolves. The details of the online SVD algorithm are discussed in the following subsection.

Online SVD

When dealing with dynamic data, updating POD or recomputing the entire SVD from scratch is computationally expensive and time-intensive. Traditional SVD also requires storing and processing the entire dataset simultaneously, further increasing computational demands. To address these challenges, researchers have demonstrated that online SVD is an effective solution.[69–72]

Online SVD initializes the decomposition using a small subset of data in the conventional manner. As new data become available, it updates existing SVD results rather than re-computing them from the beginning. This incremental approach significantly reduces computational costs and resource requirements. This sequential updating method has proven effective in various applications involving continuous data streams. By leveraging prior calculations instead of restarting the process for each update, online SVD enhances efficiency in handling evolving datasets.

Consider a matrix $\mathbf{X}_s$, which represents a subset of the response matrix $\mathbf{D}$, where $\mathbf{X}_s \in \mathbb{R}^{N \times n_k}$ and n_k denotes the number of columns used for initializing the SVD in offline mode. Notably, n_k is significantly smaller than n ($n_k \ll n$). The SVD of $\mathbf{X}_s$ is given as:

$$\mathbf{X}_s = \mathcal{U}\boldsymbol{\Lambda}\mathcal{V}^T \tag{8}$$

where $\mathcal{U} \in \mathbb{R}^{N \times N}$, $\boldsymbol{\Lambda} \in \mathbb{R}^{N \times n_k}$, and $\mathcal{V} \in \mathbb{R}^{n_k \times n_k}$.

Let $\mathbf{e} \in \mathbb{R}^{N \times 1}$ be a newly added column vector to the matrix $\mathbf{X}_s$. Consequently, the updated matrix $\mathbf{X}_u$ is given by:

$$\begin{aligned} \mathbf{X}_u &= [\mathbf{X}_s \quad \mathbf{e}] \\ &= [\mathcal{U}\boldsymbol{\Lambda}\mathcal{V}^T \quad \mathbf{e}] \end{aligned} \tag{9}$$

The core concept of online SVD is to make use of the existing SVD of the matrix $\mathbf{X}_s$ to efficiently determine the SVD of the updated matrix $\mathbf{X}_u$. In particular, whenever new data becomes available, the SVD of the updated matrix is computed by utilizing the previously obtained SVD. To accomplish this, the new snapshot $\mathbf{e}$ is projected onto the orthogonal complement of the space spanned by the columns of $\mathcal{U}$. This projection is given by:

$$\mathbf{p} = \mathbf{e} - \mathcal{U}\mathcal{U}^T\mathbf{e} \tag{10}$$

where $\mathbf{p}$ represents the projection of $\mathbf{e}$ onto the orthogonal complement of the space spanned by the columns of $\mathcal{U}$. The magnitude of $\mathbf{p}$ is denoted as h, and the normalized form of $\mathbf{p}$ is expressed as:

$$h = \|\mathbf{p}\|, \quad \mathbf{j} = \frac{\mathbf{p}}{h} \tag{11}$$

Further, Equation (9) can be rewritten in matrix form as[70]:

$$\mathbf{X}_u = [\mathcal{U} \quad \mathbf{j}] \underbrace{\begin{bmatrix} \boldsymbol{\Lambda} & \mathcal{U}^T\mathbf{e} \\ 0 & h \end{bmatrix}}_{\mathbf{E}} \begin{bmatrix} \mathcal{V} & 0 \\ 0 & 1 \end{bmatrix}^T \tag{12}$$

The SVD of $\mathbf{X}_u$ can be determined by computing the SVD of the matrix $\mathbf{E}$,[70] as shown below:

$$\mathbf{E} = \begin{bmatrix} \mathbf{\Lambda} & \mathcal{U}^T\mathbf{e} \\ 0 & h \end{bmatrix} = \mathcal{U}_E \mathbf{\Lambda}_E \mathcal{V}_E^T \tag{13}$$

Equation (13) denotes the SVD of $\mathbf{E}$. Furthermore, the SVD of $\mathbf{x}_u$ is expressed as follows:

$$\mathbf{X}_u = \underbrace{\left([\mathcal{U} \quad \mathbf{j}]\mathcal{U}_E\right)}_{\mathcal{U}_u} \mathbf{\Lambda}_E \underbrace{\left(\begin{bmatrix} \mathcal{V} & 0 \\ 0 & 1 \end{bmatrix} \mathcal{V}_E\right)^T}_{\mathcal{V}_u^T} \tag{14}$$

where $\mathcal{U}_u$ and $\mathcal{V}_u$ represent the updated left and right singular vector matrices, respectively, while $\mathbf{\Lambda}_E$ denotes the updated singular value matrix.

The procedure for computing POD bases online is described in Algorithm 1.[70,72] To reduce the impact of round-off errors, any projection h below the tolerance threshold e_h is discarded. Before initiating the update process, the SVD of the matrix $\mathbf{E}$ is computed (Algorithm 1, line $10-13$). Next, a decision is made regarding whether incorporating the snapshot column ($\mathbf{e}$) increases the rank of the updated matrix (Algorithm 1, lines 16–21). The number of columns retained in $\mathcal{U}$ is determined based on energy criteria as.

$$\frac{\sum_{j=1}^{n_a} \mathbf{\Lambda}_{E_{jj}}^2}{\sum_{j=1}^{n_k} \mathbf{\Lambda}_{E_{jj}}^2} \geqslant \epsilon_E \tag{15}$$

where ϵ_E is a constant that varies between 0 and 1. The variable n_a represents the number of required number of POM to achieve the desired level of cumulative energy.

Additionally, any singular value smaller than the predefined threshold e_{sv} is truncated to improve the efficiency of the method. Finally, the updated modes are re-orthogonalized using the modified Gram-Schmidt method to address any potential non-orthogonality that may arise among them.[70]

UKF for state and parameter estimation

The UKF has emerged as a powerful tool for state and parameter estimation in nonlinear systems, particularly for multidegree of freedom structural dynamics applications. This mathematical framework enables accurate estimation of both system states and parameters simultaneously, which proves invaluable for SHM. When applied to MDOF systems, the UKF effectively handles the complexities of coupled dynamics and nonlinearities while maintaining computational efficiency. For state and parameters estimation purpose the process equation which is a second-order differential equation defined in Equation (1) must be represented into a first-order state-space representation. This is achieved by defining the state $\mathbf{X}$ as the concatenation of displacements and velocities as:

$$\mathbf{X}(t) = \begin{bmatrix} \mathbf{x}(t) \\ \dot{\mathbf{x}}(t) \end{bmatrix} \tag{16}$$

where $\mathbf{X}(t) \in \mathbb{R}^{2N}$ contains displacement and velocity. The resulting state space model becomes:

$$\dot{\mathbf{X}}(t) = \begin{bmatrix} \dot{\mathbf{x}}(t) \\ \ddot{\mathbf{x}}(t) \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{0}_{N\times N} & \mathbf{I}_{N\times N} \\ -\mathbf{M}^{-1}\mathbf{K}(t) & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix}}_{\mathbf{A^c}} \begin{bmatrix} \mathbf{x}(t) \\ \dot{\mathbf{x}}(t) \end{bmatrix} + \underbrace{\begin{bmatrix} \mathbf{0}_{N\times N} \\ \mathbf{M}^{-1} \end{bmatrix}}_{\mathbf{B^c}} \mathbf{F}(t) \tag{17}$$

where $\mathbf{0}_{N\times N}$ and $\mathbf{I}_{N\times N}$ are null and identity matrix of size $N\times N$, respectively. $\mathbf{A^c}$, $\mathbf{B^c}$, are system and control matrix in continuous time domain, respectively. The observation equation is given as:

$$\mathbf{y}(t) = \mathbf{H}\mathbf{X}(t) \tag{18}$$

where $\mathbf{y}(t) \in \mathbb{R}^N$ is vector of N observables. $\mathbf{H} \in \mathbb{R}^{N\times 2N}$ is a Boolean matrix, which links the state matrix of the system to observation in $\mathbf{y}$.

The discretization of Equations (17) and (18) at a sampling period Δt, assuming constant intersample behavior of the excitations, is given by:

$$\mathbf{X}_{k+1} = \mathbf{A}^d\mathbf{X}_k + \mathbf{B}^d\mathbf{F}_k \tag{19}$$

$$\mathbf{y}_k = \mathbf{H}\mathbf{X}_k \tag{20}$$

The discrete-time matrices $\mathbf{A}^d$ and $\mathbf{B}^d$ are computed using the matrix exponential of the continuous-time state matrices $\mathbf{A}^c$ and $\mathbf{B}^c$ as:

$$\mathbf{A}^d = e^{\mathbf{A^c}\Delta t} \approx \mathbf{I} + \mathbf{A}^c\Delta t + 0.5\Delta t^2[\mathbf{A^c}]^2 + \ldots, \tag{21}$$

$$\mathbf{B}^d = [\mathbf{A^c} - \mathbf{I}][\mathbf{A^c}]^{-1}\mathbf{B^c} \approx \mathbf{B^c}\Delta t \tag{22}$$

To estimate the unknown parameters $\boldsymbol{\theta}$ of the system, it is necessary to incorporate them into an augmented state ($\mathbf{X^a}$), defined as:

$$\mathbf{X}_k^a = \begin{bmatrix} \mathbf{x}_k \\ \dot{\mathbf{x}}_k \\ \boldsymbol{\theta}_k \end{bmatrix} \tag{23}$$

Algorithm 1. Pseudo code online SVD

Input: $\mathcal{U} \in \mathbb{R}^{N \times n_k}, \mathbf{\Lambda} \in \mathbb{R}^{k \times n_k}, \mathcal{V} \in \mathbb{R}^{n_k \times n_k}, \mathbf{e} \in \mathbb{R}^{N \times 1}, e_h, e_{sv}$
$n_k = \text{nColumns}(\mathcal{U})$ ▷ Number of columns in $\mathcal{U}$
$h = \|\mathbf{p}\|$ ▷ Refer to Equation 11

if $h < e_h$ **then**

$$\mathbf{E} = \begin{bmatrix} \mathbf{\Lambda} & \mathcal{U}^T\mathbf{e} \\ 0 & 0 \end{bmatrix}$$

else

$$\mathbf{E} = \begin{bmatrix} \mathbf{\Lambda} & \mathcal{U}^T\mathbf{e} \\ 0 & h \end{bmatrix}$$

end if
$\mathcal{U}_E, \mathbf{\Lambda}_E, \mathcal{V}_E = \text{SVD}(\mathbf{E})$ ▷ Refer to Equation 13
if $(h < e_h)$ OR $(n_k \geqslant N)$ **then**
$\mathcal{U} = \mathcal{U}\mathcal{U}_{E(1:n_k,\,1:n_k)}$, $\mathbf{\Lambda} = \mathbf{\Lambda}_{E(1:n_k,\,1:n_k)}$, $\mathcal{V} = \begin{bmatrix} \mathcal{V} & 0 \\ 0 & 1 \end{bmatrix}\mathcal{V}_E$
else
$\mathcal{U} = [\mathcal{U} \quad \mathbf{j}]\,\mathcal{U}_E$, $\mathbf{\Lambda} = \mathbf{\Lambda}_E$, $\mathcal{V} = \begin{bmatrix} \mathcal{V} & 0 \\ 0 & 1 \end{bmatrix}\mathcal{V}_E$ ▷Refer to Equation 14
$n_k = n_k + 1$
end if
if $(n_k > n_a)$ **then** ▷ Refer to Equation 15
$\mathbf{\Lambda} = \mathbf{\Lambda}_{(1:n_a,\,1:n_a)}$
$\mathcal{U} = \mathcal{U}(:, 1 : n_a)$
$\mathcal{V} = \mathcal{V}(:, 1 : n_a)$
$n_k = n_a$
end if
if $(\mathbf{\Lambda}_{(n_k-1,\,n_k-1)} > e_{sv})$ AND $(\mathbf{\Lambda}_{(n_k,\,n_k)} < e_{sv})$ **then** ▷ Singular value truncation
$n_k = n_k - 1$
$\mathbf{\Lambda} = \mathbf{\Lambda}_{(1:n_k,\,1:n_k)}$
$\mathcal{U} = \mathcal{U}(:, 1 : n_k)$
$\mathcal{V} = \mathcal{V}(:, 1 : n_k)$
end if
if $|\mathcal{U}^T_{(:,\,n_k)}\mathcal{U}_{(:,\,1)}| > \min(e_h, \epsilon \times N)$ **then**
$\mathcal{U} = \text{ModifiedGramSchmidt}(\mathcal{U})$ ▷Reorthogonalization
end if
Output: $\mathcal{U}, \mathbf{\Lambda}, \mathcal{V}$

where $\theta \in \mathbb{R}^{n_\theta \times 1}$, is vector of unknown parameters of size n_θ. The recursive discrete time process equation can be defined by assuming zero-order-hold as[73]:

$$\mathbf{X}^a_{k+1} = g\left(\mathbf{X}^a_k\right) + \mathbf{B}_k\mathbf{F}_k + \mathbf{w}_k \tag{24}$$

where $g(\cdot)$ is a nonlinear function of augmented state that relates augmented state at k-th time instant to the $(k+1)$-th time instant $\mathbf{w} = \begin{bmatrix} \mathbf{w}_x \\ \mathbf{w}_\theta \end{bmatrix}$, is a process noise with noise covariance $\mathbf{Q}$, modeled as zero mean Gaussian with appropriate covariance value, where $\mathbf{w}_x$ and $\mathbf{w}_\theta$ are the process noise associated with the state and parameter. Nonlinear function $g(\cdot)$ and $\mathbf{B}$ can be derived with Equation (17) as:

$$g\left(\mathbf{X}^a_k\right) = \begin{bmatrix} \mathbf{A}^d_k & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}\mathbf{X}^a_k, \qquad \mathbf{B}_k = \begin{bmatrix} \mathbf{B}^d_k \\ \mathbf{0} \end{bmatrix} \tag{25}$$

The observation equation at k-th time instant is written as:

$$\mathbf{y}_k = h\left(\mathbf{X}^a_k\right) + \mathbf{v_k} \tag{26}$$

where $\mathbf{v_k}$ is the process noise with covariance $\mathbf{R}$ modeled as zero mean Gaussian with appropriate covariance value, and function $h(\cdot)$ is given as:

$$h\left(\mathbf{X}^a_k\right) = \mathcal{H}\mathbf{X}^a_k \tag{27}$$

where $\mathcal{H} = [\mathbf{H} \quad \mathbf{0}]$, selects the measured DOFs.

The UKF employs the UT to compute the statistical moments of states undergoing a nonlinear transformation. The UT uses a deterministic sampling technique to capture the first (mean) and second (covariance) statistical moments through a carefully chosen set of sample points, known as sigma points. For an l-dimensional augmented state vector $\mathbf{X}^a$ with mean $\overline{\mathbf{X}}^a$ and state estimation error $\mathbf{P}$ representing the

uncertainty associated with the state equation. The set of $2l+1$ sigma points, $\mathcal{X}$, is computed as:

$$\begin{aligned}\mathcal{X}_0 &= \overline{\mathbf{X}}^a,\\ \mathcal{X}_i &= \overline{\mathbf{X}}^a + (\sqrt{(l+\lambda)\mathbf{P}})_i, \quad i=1,\ldots,l,\\ \mathcal{X}_{i+l} &= \overline{\mathbf{X}}^a - (\sqrt{(l+\lambda)\mathbf{P}})_i, \quad i=1,\ldots,l.\end{aligned} \tag{28}$$

where $\left(\sqrt{(l+\lambda)\mathbf{P}}\right)_i$ represents the i-th column of the matrix square root of $(l+\lambda)\mathbf{P}$, and λ is a scaling parameter defined as $\lambda = \alpha^2(l+\kappa) - l$. α is a parameter that controls the spread of sigma points around the mean, and κ is a secondary scaling parameter, often set to 0 or $3-l$. The square root of the covariance matrix can be computed using methods such as Cholesky decomposition or QR decomposition.

Each sigma point is associated with a weight used to compute the mean and covariance of the transformed distribution. These weights are calculated as:

$$\begin{aligned}W_0^m &= \frac{\lambda}{l+\lambda},\\ W_0^c &= \frac{\lambda}{l+\lambda} + \left(1-\alpha^2+\beta\right),\\ W_i^m &= W_i^c = \frac{1}{2(l+\lambda)}, \quad i=1,\ldots,2l.\end{aligned} \tag{29}$$

where W_i^m denotes the weights for calculating the mean, W_i^c represents the weights for calculating the covariance, and β is a parameter that incorporates prior knowledge of the distribution.

The UKF propagates sigma points through the nonlinear process model to predict the next state and its associated covariance. The sigma points at k-th time are generated as:

$$\mathcal{X}_k = \left[\overline{\mathbf{X}}_k^a \quad \overline{\mathbf{X}}_k^a + \sqrt{(l+\lambda)\mathbf{P}_k} \quad \overline{\mathbf{X}}_k^a - \sqrt{(l+\lambda)\mathbf{P}_k}\right] \tag{30}$$

These sigma points are then propagated through the nonlinear system dynamics as defined in Equation (24):

$$\mathcal{X}_k^x = g(\mathcal{X}_k) + \mathbf{B}_k\mathbf{F}_k \tag{31}$$

where $\mathcal{X}_k^x$ represents the state computed at the k-th time step at the sigma points $\mathcal{X}$.

The predicted state mean is obtained by computing the weighted sum:

$$\hat{\overline{\mathbf{X}}}_k = \sum_{i=0}^{2n} W_i^m \mathcal{X}_{i,k}^x \tag{32}$$

Similarly, the predicted state covariance is computed as:

$$\mathbf{P}_k = \sum_{i=0}^{2n} W_i^c [\mathcal{X}_{i,k}^x - \hat{\overline{\mathbf{X}}}_k][\mathcal{X}_{i,k}^x - \hat{\overline{\mathbf{X}}}_k]^T + \mathbf{Q}_k \tag{33}$$

After predicting the state, the UKF updates the estimate using the available measurements. The measurement sigma points are computed from Equation (26) as:

$$\mathcal{Y}_k = h\left[\mathcal{X}_k^x\right] \tag{34}$$

Mean and covariance for $\mathbf{y}$ are approximated using a weighted sample mean and covariance of the posterior sigma points as:

$$\hat{\overline{\mathbf{y}}}_k = \sum_{i=0}^{2n} W_i^m \mathcal{Y}_{i,k} \tag{35}$$

The measurement covariance (also called the innovation covariance) is:

$$\mathbf{P}_{y_k y_k} = \sum_{i=0}^{2l} W_i^c \left[\mathcal{Y}_{i,k} - \hat{\overline{\mathbf{y}}}_k\right]\left[\mathcal{Y}_{i,k} - \hat{\overline{\mathbf{y}}}_k\right]^T + \mathbf{R_k} \tag{36}$$

The cross-covariance between the state and measurement is computed as:

$$\mathbf{P}_{x_k y_k} = \sum_{i=0}^{2l} W_i^c [\mathcal{X}_{i,k}^x - \hat{\overline{\mathbf{X}}}_k]\left[\mathcal{Y}_{i,k} - \hat{\overline{\mathbf{y}}}_k\right]^T \tag{37}$$

Kalman gain is computed by using Equations (36) and (37) as:

$$\mathbf{G}_k = \mathbf{P}_{x_k y_k}\mathbf{P}_{y_k y_k}^{-1} \tag{38}$$

Finally, the state and covariance are updated as follows:

$$\hat{\mathbf{X}}_k = \hat{\overline{\mathbf{X}}}_k + \mathbf{G}\left(\mathbf{y}_k - \hat{\overline{\mathbf{y}}}_k\right) \tag{39}$$

$$\mathbf{P}_k = \mathbf{P}_k - \mathbf{G}_k\mathbf{P}_{y_k y_k}\mathbf{G}_k^T \tag{40}$$

Pseudocode for UKF is presented in Algorithm 2.

POD-assisted UKF algorithm

This study proposes a novel methodology that integrates POD with the UKF for real-time damage localization and quantification in dynamic systems. The

Algorithm 2. Unscented Kalman Filter

Initialize:
$\hat{\mathbf{X}}_0 = \mathbb{E}[\mathbf{X}_0]$
$\mathbf{P}_0 = \mathbb{E}[(\mathbf{X}_0 - \hat{\mathbf{X}}_0)\mathbf{X}_0 - \hat{\mathbf{X}}_0)^T]$
$\mathbf{Q}$ = process noise covariance
$\mathbf{R}$ = measurement noise covariance
for $k = 1, 2, \ldots$ **do**

$\mathcal{X}_k = \left[\overline{\mathbf{X}}_k^a \quad \overline{\mathbf{X}}_k^a + \sqrt{(l+\lambda)\mathbf{P}_k} \quad \overline{\mathbf{X}}_k^a - \sqrt{(l+\lambda)\mathbf{P}_k}\right]$ ▷ Refer Equation 30

$\mathcal{X}_k^x = g(\mathcal{X}_k, \mathbf{F}_k)$ ▷ Refer Equation 31

$\hat{\overline{\mathbf{X}}}_k = \sum_{i=0}^{2n} W_i^m \mathcal{X}_{i,k}^x$ ▷ Refer Equation 32

$\mathbf{P}_k = \sum_{i=0}^{2n} W_i^c [\mathcal{X}_{i,k}^x - \hat{\overline{\mathbf{X}}}_k][\mathcal{X}_{i,k}^x - \hat{\overline{\mathbf{X}}}_k]^T + \mathbf{Q}_k$ ▷ Refer Equation 33

$\mathcal{Y}_k = h\left[\mathcal{X}_k^x\right]$ ▷ Refer Equation 34

$\hat{\overline{\mathbf{y}}}_k = \sum_{i=0}^{2n} W_i^m \mathcal{Y}_{i,k}$ ▷ Refer Equation 35

$\mathbf{P}_{y_k y_k} = \sum_{i=0}^{2n} W_i^c \left[\mathcal{Y}_{i,k} - \hat{\overline{\mathbf{y}}}_k\right]\left[\mathcal{Y}_{i,k} - \hat{\overline{\mathbf{y}}}_k\right]^T + \mathbf{R}_k$ ▷ Refer Equation 36

$\mathbf{P}_{x_k y_k} = \sum_{i=0}^{2n} W_i^c)[\mathcal{X}_{i,k}^x - \hat{\overline{\mathbf{X}}}_k]\left[\mathcal{Y}_{i,k} - \hat{\overline{\mathbf{y}}}_k\right]^T$ ▷ Refer Equation 37

$\mathbf{G}_k = \mathbf{P}_{x_k y_k} \mathbf{P}_{y_k y_k}^{-1}$ ▷ Refer Equation 38

$\hat{\mathbf{X}}_k = \hat{\overline{\mathbf{X}}}_k + \mathbf{G}_k\left(\mathbf{y}_k - \hat{\overline{\mathbf{y}}}_k\right)$ ▷ Refer Equation 39

$\mathbf{P}_k = \mathbf{P}_k - \mathbf{G}_k \mathbf{P}_{y_k y_k} \mathbf{G}_k^T$ ▷ Refer Equation 40
end for

overall flow of the proposed approach is illustrated in Figure 1. A healthy POM slope is considered as the baseline and can be computed using either numerically simulated responses or experimental data.

Although POD is generally considered insensitive to loading conditions,[63] minor variations in the POMs can still occur due to changes in excitation and system uncertainties. To account for this variability, a number of simulations are conducted under different realizations of zero-mean white Gaussian excitation with unit covariance. For each simulation, the first POM is extracted and its slope is computed. The baseline POM slope ($\dot{\phi}$) is then defined as the mean of all computed POM slopes plus four times their standard deviation. This statistically informed threshold ensures a conservative and robust reference, allowing clear differentiation between normal structural variability and changes indicative of damage.

In the online setting, POD is employed to compute POMs and compute its spatial derivative in real time using an online SVD approach. In this study, only the first POM is used, and references to POMs henceforth refer specifically to this mode unless otherwise stated.

Damage localization is achieved by comparing the evolving POM slope ($\dot{\mathcal{U}}$) during system operation with the established baseline. Deviations beyond the baseline indicate potential damage, and the associated DOFs are flagged as affected as shown in Figure 1. The UKF operates in parallel with the online SVD, initially estimating only the structural states (e.g., displacements and velocities). Once damage is detected, the UKF dynamically augments its state vector to include stiffness parameters corresponding only to the affected DOFs. This selective augmentation avoids unnecessary parameter estimation across all DOFs, thereby reducing computational burden and maintaining numerical stability. The state dimension increases adaptively based on the number of damaged locations detected in real time. This adaptive estimation strategy offers significant advantages over conventional approaches. The proposed framework significantly reduces computational cost by avoiding full-parameter estimation. Its real time capability facilitates continuous monitoring and timely detection of structural changes. The combined use of online SVD for damage localization and UKF for selective parameter estimation ensures

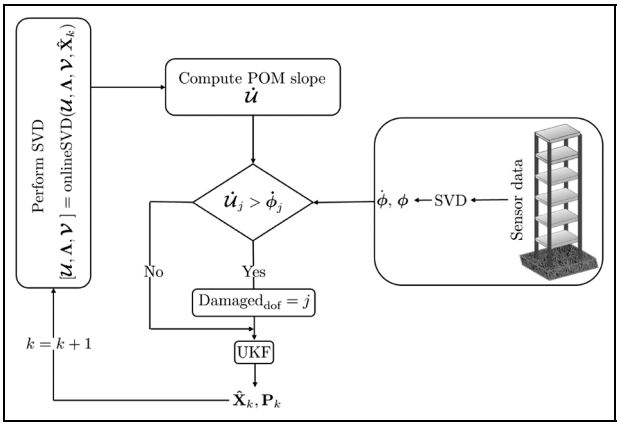


Figure 1. Schematic workflow of the proposed POD-assisted Bayesian estimation.
POD: proper orthogonal decomposition.

efficient and accurate tracking of the structural state. This approach effectively addresses three out of the four standard stages of SHM: detection, localization, and quantification excluding only the prediction of remaining useful life. By focusing estimation efforts solely on the affected DOFs, the framework strikes a compelling balance between computational efficiency and accuracy, rendering it highly suitable for real time SHM applications. Furthermore, to clarify the efficiency of the proposed method, we report the computational effort of the two main components of the framework in the Appendix section. Although both components, (i) the online SVD for updating the POMs and (ii) the UKF for estimating state and stiffness parameters, operate simultaneously within the proposed framework, they contribute to the total cost in different ways. The POMs update requires repeated decomposition of the response data; therefore, a comparison of the cost of full SVD with the online SVD is presented. The full SVD becomes increasingly expensive as the response matrix grows, whereas the online SVD updates the basis using only the newest snapshot and avoids storing past data. The second component is the state and parameter-estimation step. The computational cost associated with the classical UKF and the POD-assisted UKF is presented. The computational cost of the UKF increases cubically with the number of parameters being estimated. The classical UKF updates all stiffness parameters, while the POD-assisted updates only those associated with the detected damage. This leads to a smaller augmented state and reduction in computation cost.

Numerical study

A numerical study is conducted using an eight-story shear building benchmark problem,[74] as illustrated in Figure 2(a). The structure consists of eight stories, each with a floor mass of 625×10^3 kg and an interstory stiffness of $10^6\,\text{kN}\,\text{m}^{-1}$. To incorporate damping effects, 2% Rayleigh damping is assigned to the structure. To establish a reference POM under healthy conditions, a

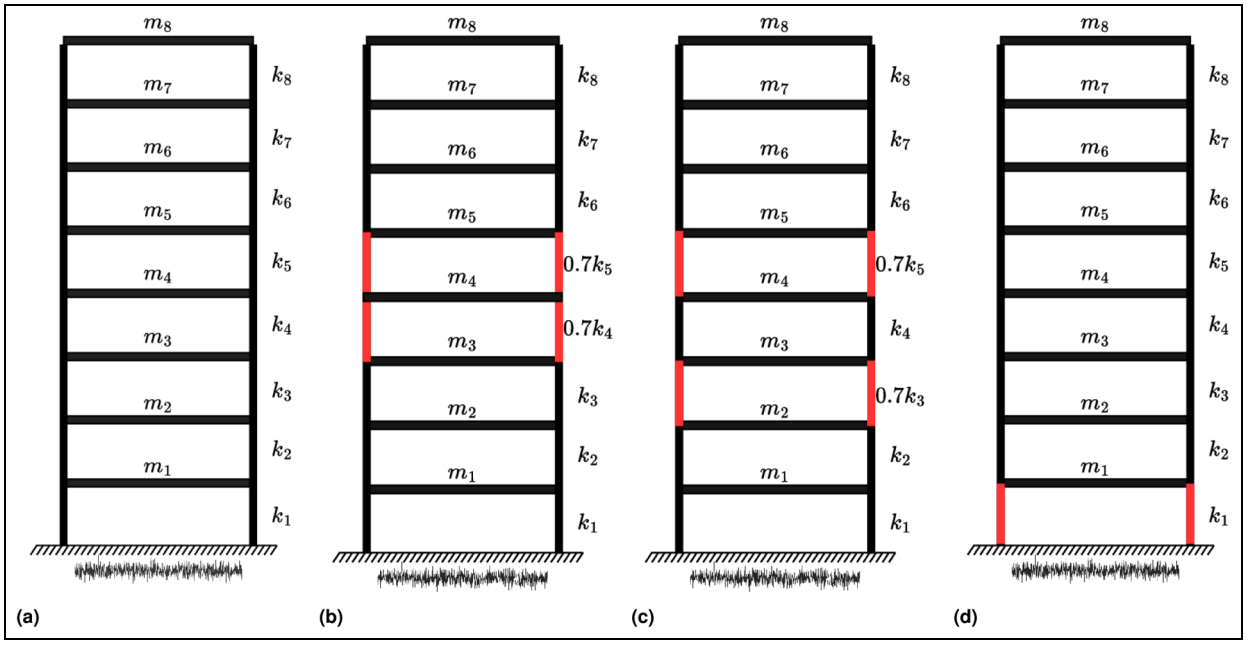


Figure 2. An eight-story building with base excitation: (a) Healthy, (b) 30% damage at fourth–fifth stories, (c) third–fifth stories, and (d) first story.

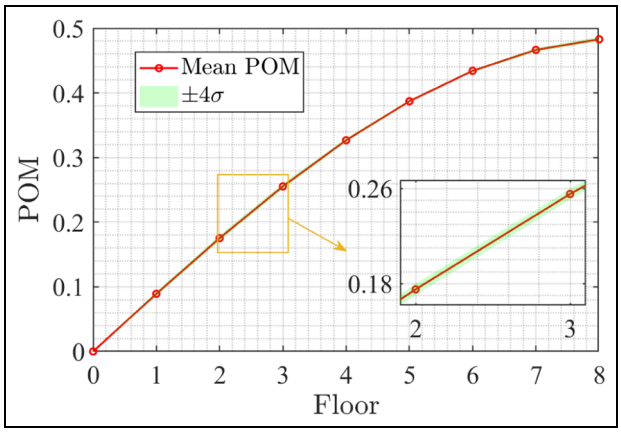


Figure 3. First POM with mean and standard deviation bound.
POM: proper orthogonal mode.

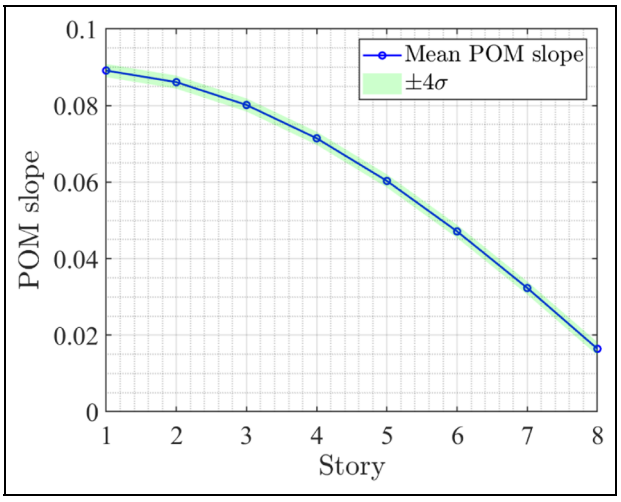


Figure 4. First POM slope with mean and standard deviation bound.
POM: proper orthogonal mode.

total of 1×10^4 numerical simulations are performed. In each simulation, the structure retains identical physical properties, while the base excitation varies across realizations. The excitation is modeled as zero-mean white Gaussian noise with unit covariance. Structure is subjected to this excitation for 10 s with a time step of 0.001 s, ensuring a high temporal resolution for capturing dynamic response variations. This ensemble of simulations provides insight into the variation of POMs arising from different excitation forces. Further, SVD is employed to the displacement response data to extract the POM. The POM, along with its mean and ±4 standard deviation (σ) bounds, is shown in Figure 3. Once the POMs are computed, their mean slopes, along with the corresponding $\pm4\sigma$ bounds, are also evaluated and shown in Figure 4. Since any structural damage would alter the stiffness distribution, it would also affect the POM and its slope. Thus, deviations in the slope of the POM from its baseline behavior serve as a reliable indicator of structural damage, enabling early detection of changes in the dynamic properties of the system.

Damage is simulated by reducing the stiffness of specific stories in the structure. Three distinct damage scenarios are considered to assess the effectiveness of the proposed framework and are summarized in Table 1.

The UKF parameters, α, β, and κ, are assigned values of 1×10^{-3}, 2, and 0, respectively.[55] The process noise covariance is set to 10^{-11}, while the measurement noise covariance is set to 10^{-6}. Additionally, the process noise covariance for the parameters is set to 10^{10}. It is noteworthy that these UKF parameters remain the same across all cases considered. In the following subsection, each damage case is discussed in detail.

Table 1. Considered damage cases.

Damaged case	Story number	Damage	Damage instance
Case 1	4 & 5	30%	50 s
Case 2	3 & 5	30%	50 s
Case 3	1	15%, 30%, 50%	25 s, 80 s, 125 s

Damaged case 1

In this case, the building is subjected to zero-mean white Gaussian base excitation with unit variance for a duration of 200 s. The structural response is computed using numerical integration with a time step of 0.001 s. To simulate realistic measurement, white Gaussian noise is added to the displacement response, which has made the response data noisy. The variance of the added noise is taken as 5% of the variance of the displacement signal. Double damage is introduced by reducing the stiffness of two consecutive stories (fourth and fifth) by 30% at 0.001 s, as shown in Figure 2(b). This scenario is considered to evaluate the framework's ability to detect and distinguish between multiple adjacent damage locations.

In Figure 5, the temporal evolution of the POM slope is presented. The shaded black surface represents the baseline POM slope, while the colored surface depicts the real-time evolution of the POM slope during the monitoring period. It is clear that the POM slope begins to change after the damage occurs at

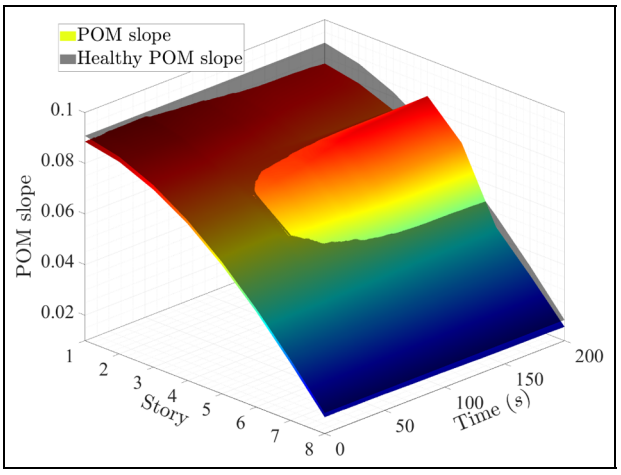


Figure 5. Evolution of POM slope over time for damage case 1. POM: proper orthogonal mode.

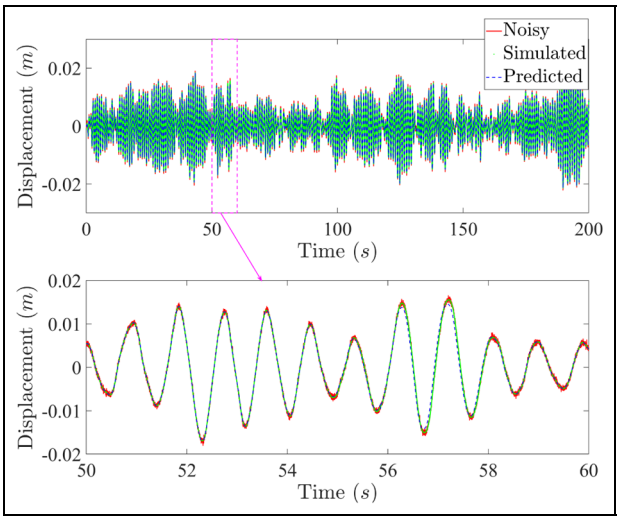


Figure 6. Comparison of predicted and simulated displacement at the fifth floor level for damage case 1. Top: Full-time comparison. Bottom: Zoomed-in view between 50 s and 60 s highlighting the damage detection period.

$t = 50$ s. However, it took approximately 5 s for the POM values at the damaged locations to rise above the baseline POM values. This delay is attributed to the nature of the online SVD approach, where the POM is incrementally updated as more measurement data becomes available, allowing the system to capture the structural changes caused by the damage in real time. Figures 6 and 7 show the predicted displacement and velocity responses at the fifth story. As illustrated, the predicted displacement and velocity closely matches the simulated response, highlighting the high accuracy of the proposed estimation framework. Although the

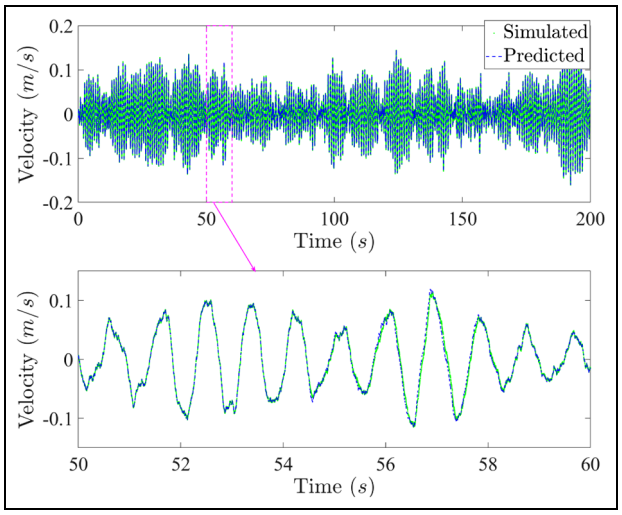


Figure 7. Comparison of predicted and simulated velocity at the fifth floor level for damage case 1. Top: Full-time comparison for damage case 1. Bottom: Zoomed-in view between 50 s and 60 s highlighting the damage detection period.

algorithm identified the structural damage approximately 5 s after it occurs at $t = 50$ s, this slight delay has minimal to no effect on the accuracy of the predicted responses. This robustness can be attributed to the capabilities of the UKF, which effectively combines model dynamics and measurement data to generate reliable state estimates, even as the structure undergoes changes. The observed correlation between the predicted and actual responses at the damaged fifth story further emphasizes the adaptability and reliability of the proposed framework in real-time structural monitoring. For brevity, results from other stories are not shown, though they exhibit similar trends. The predicted stiffness at the fourth and fifth stories is shown in Figures 8 and 9, respectively. It is evident that stiffness prediction does not begin at the initial time step but is triggered shortly after the occurrence of damage. There is an approximate delay of 5 s between the actual damage instant ($t = 50$ s) and the initiation of stiffness estimation. This delay is consistent with the POM slope evolution as shown in Figure 5, where the slope corresponding to the fourth and fifth stories begins to change immediately after the damage but takes a few seconds to fully develop and become distinguishable from the baseline.

As outlined in the methodology section, the framework omits the need to estimate all stiffness parameters continuously. Instead, it initiates stiffness prediction only for the affected stories and only after significant deviation is detected making the approach both efficient and focused. Once estimation begins, the

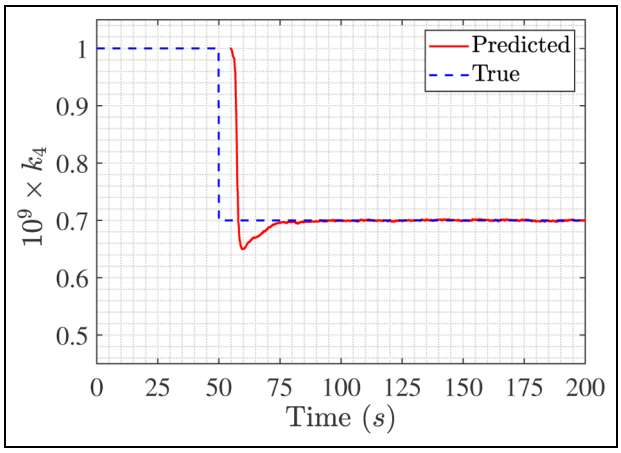


Figure 8. Comparison of true and predicted stiffness at the fourth story for damage case 1.

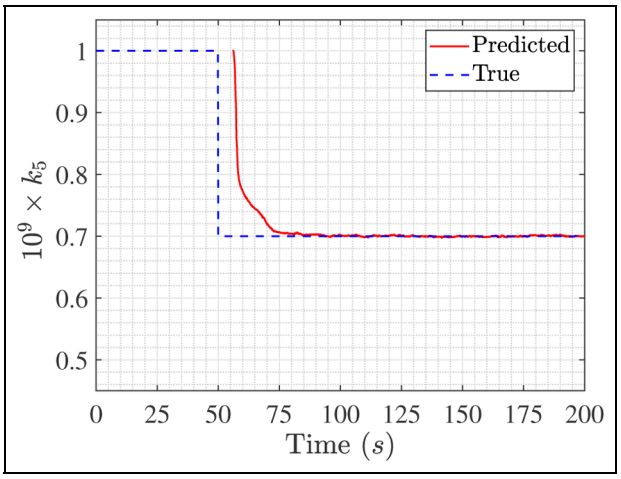


Figure 9. Comparison of true and predicted stiffness at the fifth story for damage case 1.

predicted stiffness values closely match the true stiffness, confirming the framework's capability to accurately track localized damage.

Damaged case 2

This scenario considers a non-consecutive double damage condition. The structure is excited using a zero-mean white Gaussian base acceleration with unit variance for a total duration of 200 s. Numerical integration is performed using a fixed time step of 0.001 s to compute the structural response. To replicate realistic measurement conditions, the displacement response is corrupted with white Gaussian noise. The noise variance is set to 5% of the displacement signal's variance, ensuring the simulation reflects practical sensor noise levels. Furthermore, damage is introduced by reducing the stiffness of the third and fifth stories by 30% at 50 s, as illustrated in Figure 2(c). Unlike the first case, the damage locations are spatially separated, allowing assessment of the framework's ability to detect and isolate damage when it occurs at non-adjacent stories.

In Figure 10, the temporal evolution of the POM slope is shown. The shaded black surface represents the baseline POM slope, while the colored surface illustrates real-time evolution of POM slope over the time. A clear deviation in the POM slope starts to appear shortly after the damage occurs at $t = 50$ s. Specifically, the slope values for the third and fifth stories, where the stiffness was reduced, exhibit a consistent upward trend as time progresses, confirming the presence of damage at these locations. Figures 11 and 12 present the predicted displacement and velocity responses at the fifth story, respectively. As observed, the predicted responses closely follow the simulated results, indicating the high accuracy and robustness of the proposed estimation framework. Although a delay of approximately 5 s is observed in damage detection relative to the actual occurrence at $t = 50$ s, this slight lag has negligible effect on the overall accuracy of the predicted structural response. For brevity, results from other stories are not included, although they exhibit similar behavior. The estimated stiffness for third and fifth stories is shown in Figures 13 and 14, respectively. It is important to note that the estimation process does not begin at the start of the monitoring period. Rather, it is initiated a few seconds after the damage occurs. A delay of approximately 7 s is observed between the moment of damage and the start of stiffness tracking. This delay corresponds to the evolution of the POM slope in Figure 10, where initial shifts appear after damage but take a few seconds to clearly diverge from the baseline. The online SVD's incremental updates require this short period of post-damage data to capture stable changes in system behavior.

Damaged case 3

The third scenario simulates stepwise damage at the first story, with k_1 being progressively reduced by 15% at 25 s, 30% at 80 s, and 50% at 125 s, as shown in Figure 2(d). This scenario is particularly critical, as the ground floor experiences the highest shear forces during base excitation. The structure is excited using a zero-mean white Gaussian base acceleration with unit variance for a total duration of 200 s. Numerical integration is performed using a fixed time step of 0.001 s to compute the structural response. To replicate realistic measurement conditions, the displacement response is corrupted with white Gaussian noise. The noise variance is set to 5% of the displacement signal's variance,

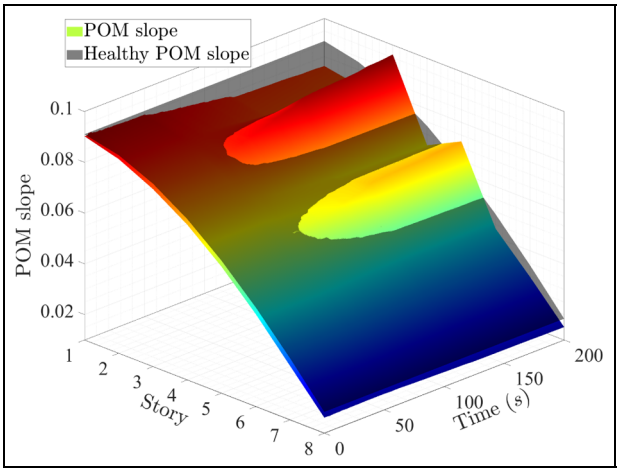


Figure 10. Evolution of POM slope over time for damage case 2. POM: proper orthogonal mode.

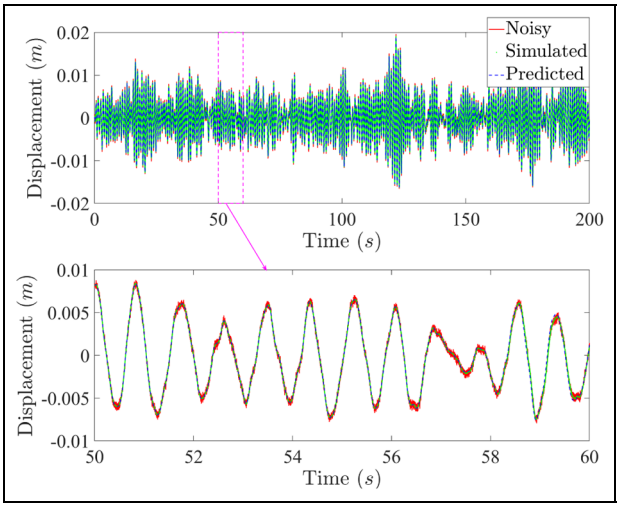


Figure 11. Comparison of predicted and simulated displacement at the fifth floor level for damage case 2. Top: Full-time comparison. Bottom: Zoomed-in view between 50 s and 60 s highlighting the damage detection period.

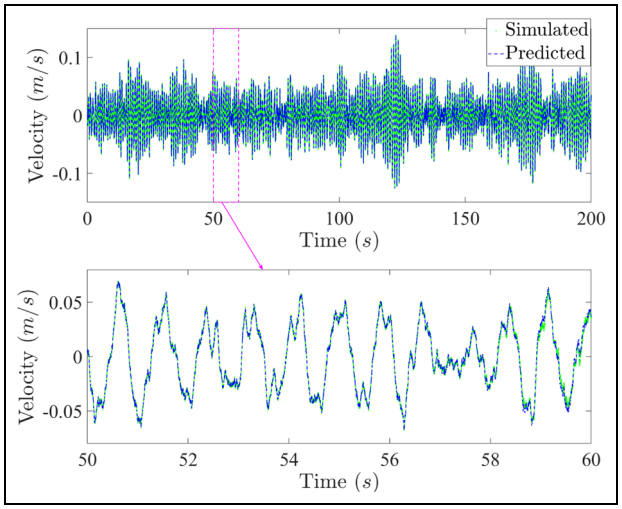


Figure 12. Comparison of predicted and simulated velocity at the fifth floor level for damage case 2. Top: Full-time comparison. Bottom: Zoomed-in view between 50 s and 60 s highlighting the damage detection period.

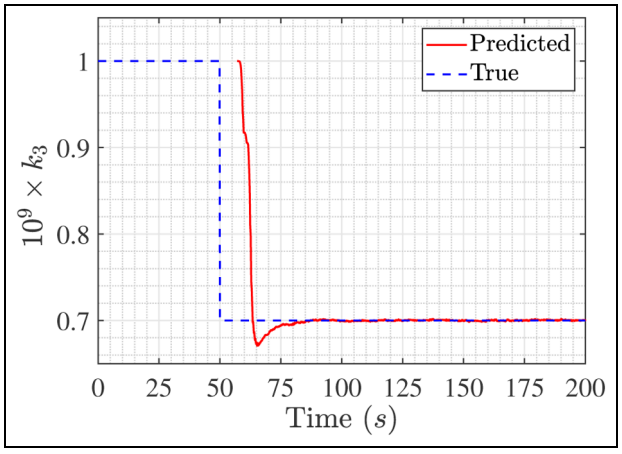


Figure 13. Comparison of true and predicted stiffness at the third story for damage case 2.

ensuring the simulation reflects practical sensor noise levels.

Figure 15 illustrates the temporal evolution of the POM slope. The black shaded surface represents the reference POM slope, while the colored surface shows POM slope variation over the time. A noticeable shift in the POM slope begins shortly after the first damage instant occurred at $t = 25$ s. Specifically, the slope values for the first story, where stiffness was reduced, exhibit a steady increase over time. Notably, the POM slope starts changing its shape around the first damage instant at 25 s, stabilizing at a nearly constant value from 40 s until the second damage instant at 80 s. A similar trend is observed for the third damage instance, indicating that the POM slope values are highly dependent on the damage level. Figures 16 and 17 show the predicted displacement and velocity responses at the first story. The predicted responses align closely with the simulated response, underscoring the reliability of the proposed estimation framework. The close match between the predicted and actual responses highlights the robustness and reliability of the proposed

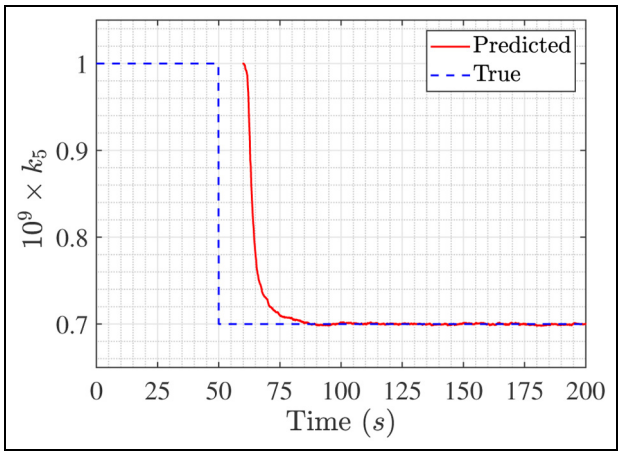


Figure 14. Comparison of true and predicted stiffness at the fifth story for damage case 2.

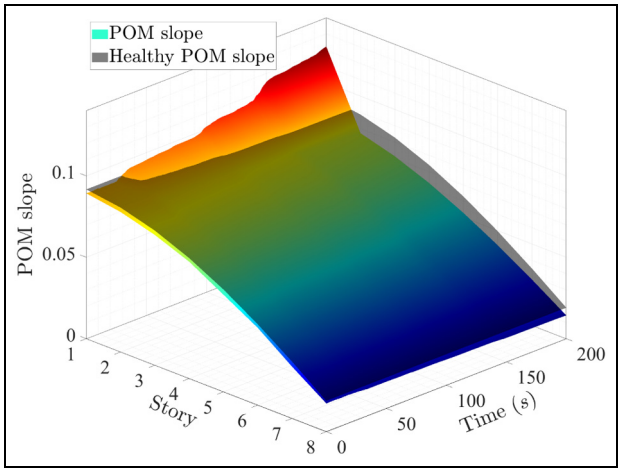


Figure 15. Evolution of POM slope over time for damage case 3. POM: proper orthogonal mode.

framework for real-time structural monitoring. For brevity, results from other stories are not included, as they follow similar trends.

Figure 18 depicts the estimated stiffness for the first story. The blue line represents the actual stiffness, while the red line shows the stiffness prediction over time. It is important to note that the estimation process does not start from the initial time but is triggered a few seconds after the damage occurs. A delay of approximately 5 s is observed between the onset of damage and when the stiffness tracking starts. This delay aligns with the evolution of the POM slope in Figure 15, where minor deviations appear shortly after the first damage instant but take a few seconds to clearly diverge from the baseline. This behavior is attributed to the incremental updates in the online

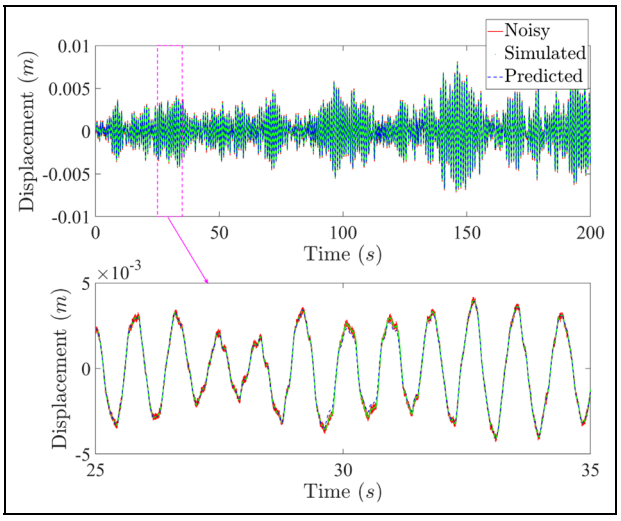


Figure 16. Comparison of predicted and simulated displacement at the first floor level for damage case 3. Top: Full-time comparison. Bottom: Zoomed-in view between 25 s and 35 s.

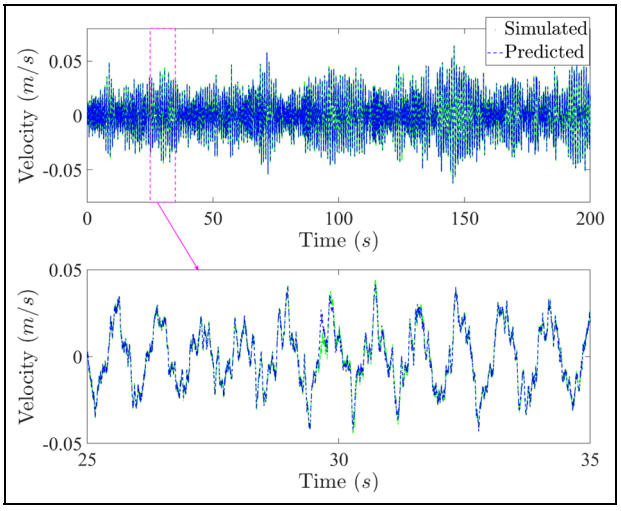


Figure 17. Comparison of predicted and simulated velocity at the first floor level for damage case 3. Top: Full-time comparison. Bottom: Zoomed-in view between 25 s and 35 s

SVD process, which require a brief period of post-damage data to capture stable variations in the system's dynamic properties. However, the further reduction in stiffness due to damage is captured accurately. The evolution of cumulative CPU time for the full SVD (using the econ option in MATLAB) and the online SVD is shown in Figure 19. The econ option is used because it computes only the economy-sized SVD, avoiding the unnecessary full factorization and significantly reducing the computation. The full SVD

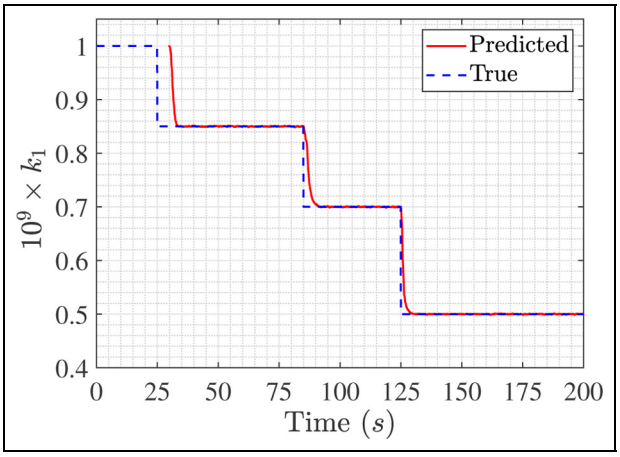


Figure 18. Comparison of true and predicted stiffness for damage case 3.

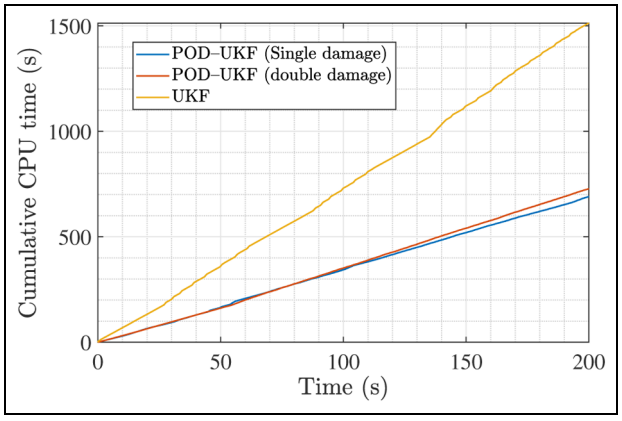


Figure 20. Cumulative CPU time for the proposed framework versus the full UKF. Updating only damaged parameters yields substantially lower computational cost.
UKF: Unscented Kalman Filter.

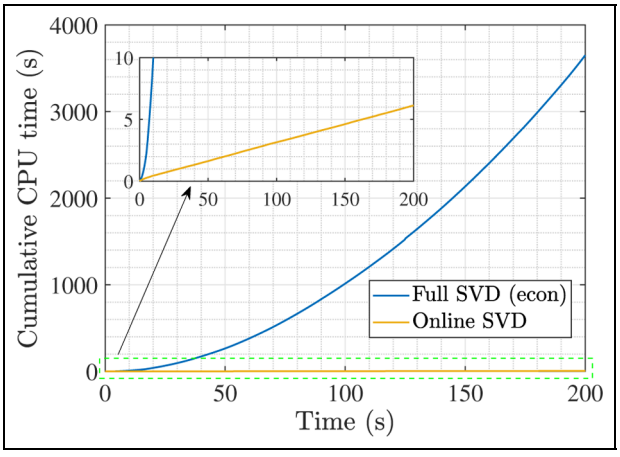


Figure 19. Comparison of cumulative CPU time: full SVD shows quadratic growth, while online SVD exhibits substantially lower, linear growth.
SVD: singular value decomposition.

requires almost 3700 s, while the online SVD completes the same updates in about 6 s. Furthermore, Figure 20 presents the cumulative CPU time comparison between the proposed POD-assisted UKF and the classical UKF. In the single-damage and double-damage cases, the proposed framework required about 690 s and 728 s, respectively, whereas the full UKF which updates stiffness parameters at all DOFs required nearly 1515 s. This demonstrates that the proposed approach reduces the computational effort by more than half, while maintaining accuracy in damage localization and parameter estimation. These CPU times vary from machine to machine, as they depend on the underlying processor, memory, and overall system configuration. All computations in this study were carried out in MATLAB R2024b on a HP laptop equipped with an 1.6 GHz Intel Core i5 and 8 GB RAM.

A three-dimensional steel building

This study utilizes experimental data obtained from a previously published investigation involving a small scale, six-story, three-dimensional steel shear building model subjected to free vibration testing.[75,76] The experimental program was conducted at the Structural Health Monitoring Laboratory of the Indian Institute of Technology Patna. The physical model was constructed at a geometric scale of 1 : 20 using mild steel, with a Young's modulus of $210\times10^9\,\mathrm{N\,m^{-2}}$, a mass density of $2.7\,\mathrm{g\,cm^{-3}}$, and a Poisson's ratio of 0.3. Each story measured 187.5 mm in height, resulting in a total structural height of 1125 mm. The floor plan measured 250 mm×250 mm. In the undamaged configuration, the columns at all stories measured 7.8 mm×7.8 mm in cross-section. To simulate damage, the columns between the third and fourth stories were replaced with smaller sections measuring 6.5 mm×6.5 mm, corresponding to an estimated stiffness reduction of approximately 51.7% at the fourth story. Free vibration tests were conducted by applying a lateral displacement to the structure and allowing it to oscillate freely. Horizontal acceleration responses were recorded at each floor using six DeltaTron piezoelectric

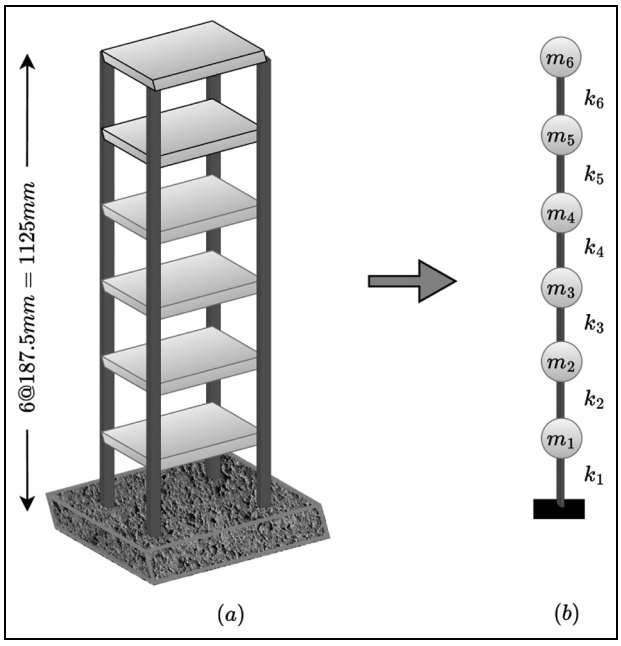


Figure 21. Representation of a six-story building and its equivalent lumped mass model: (a) Three-dimensional structural model of the building and (b) simplified lumped mass-spring representation.

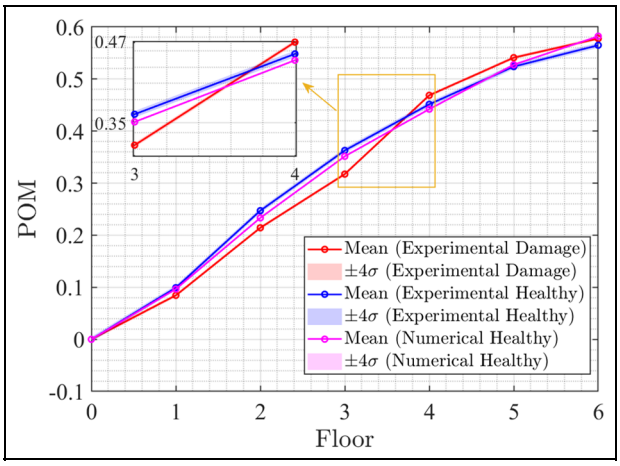


Figure 22. First POM with mean and standard deviation bound.
POM: proper orthogonal mode.

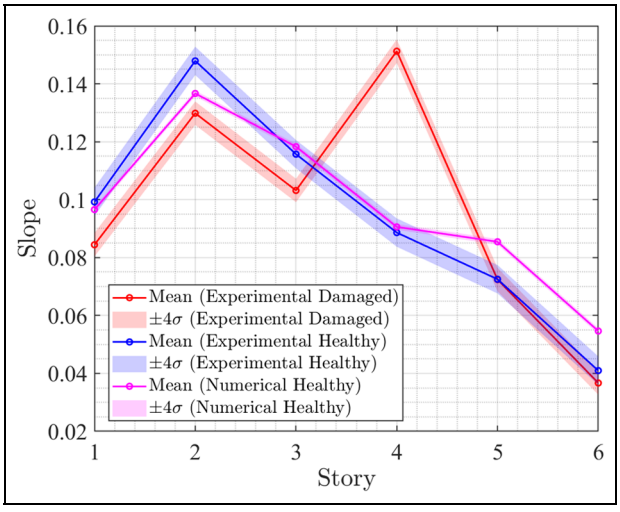


Figure 23. First POM slope with mean and standard deviation bound.
POM: proper orthogonal mode.

accelerometers oriented in the direction of motion. The sensors were connected to a multi-channel data acquisition system, and the signals were processed using Pulse Labshop and Pulse Reflex software. Data were collected at a sampling interval of 0.00024 s. A total of 25 acceleration time histories were recorded for each structural condition (healthy and damaged).[75,76]

In this study, the original three-dimensional laboratory building model is simplified into a one-dimensional lumped mass system with six DOF, where each mass corresponds to an individual floor level as shown in Figure 21. The first natural frequency and the associated mode shape, extracted from free vibration experiments, are used as reference modal parameters for calibrating the simplified model. A multi-objective genetic algorithm is employed to estimate the lateral stiffness values of each story. The optimization process minimizes the discrepancies between the numerical and experimental results in terms of both the first natural frequency and the first mode shape.

Following calibration, a numerical study is performed on the calibrated model to compute the baseline POM. A total of 10^4 simulations is performed, where the structure is subjected to zero-mean white Gaussian excitation with unit variance. The resulting POMs are compared with those obtained from the experimental data, including a confidence envelope defined by a standard deviation bound ($\pm 4\sigma$), as shown in Figure 22. As shown in Figure 22, the numerical and experimental POMs under the undamaged condition show close agreement, with minimal discrepancies. The POM corresponding to the damaged condition exhibits a noticeable kink between the third and fourth stories, aligning with the damaged location. This localized irregularity becomes even more pronounced in the slope of the POM (Figure 23), making it an effective indicator of damage. Although small differences are observed in the healthy POM slopes between the numerical and experimental cases, the damaged POM slope displays a significant deviation precisely at the fourth story, clearly confirming the

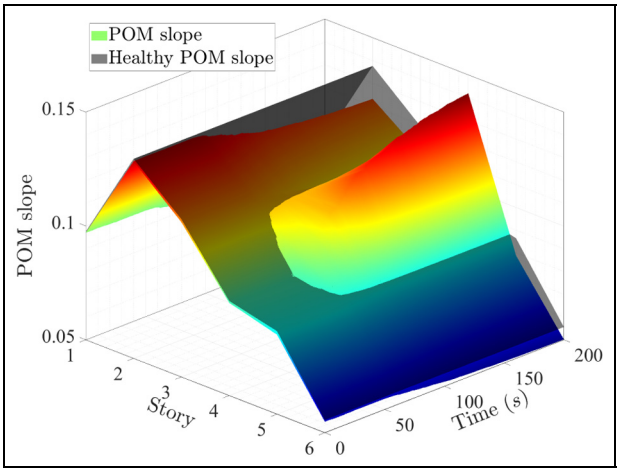


Figure 24. Evolution of POM slope over time for a three-dimensional building frame.
POM: proper orthogonal mode.

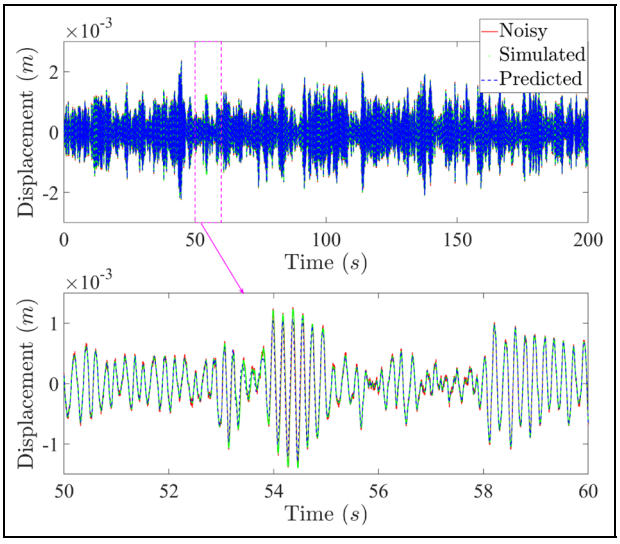


Figure 25. Comparison of predicted and simulated displacement at the fourth floor level for a three-dimensional building. Top: Full-time comparison. Bottom Zoomed-in view between 50 s and 60 s highlighting the damage detection period.

presence of damage. These observations reinforce the capability of the POM slope as a sensitive damage indicator.

To validate the proposed framework, numerical simulation is performed on the calibrated model. System responses are computed over a duration of 200 s with a time step of 0.001 s. Damage is introduced by reducing the stiffness of the fourth story by 50% at

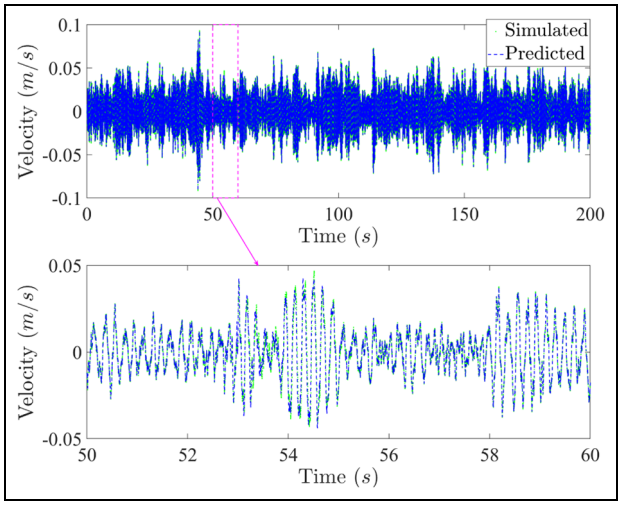


Figure 26. Comparison of predicted and simulated velocity at the fourth floor level for a three-dimensional steel building. Top: Full-time comparison. Bottom: Zoomed-in view between 50 s and 60 s highlighting the damage detection period.

50 s. The parameters for the UKF are set as $\alpha = 1\times10^{-3}$, $\beta = 2$, and $\kappa = 0$. The process noise covariance is defined as 10^{-11}, and the measurement noise covariance is set to 10^{-6}. For stiffness parameter estimation, the noise covariance is set to 10^{5} to allow for adaptive tracking of abrupt changes.

The temporal evolution of the POM slope is shown in Figure 24. The colored surface represents the real-time evolution of the POM slope, while the shaded black surface corresponds to the baseline slope under undamaged conditions. As shown in Figure 24, the real-time POM slope closely follows the baseline until the damage instant ($t = 50$ s). After this point, a sharp deviation is observed at the fourth story, indicating the structural damage. Further validation of the framework is presented in Figure 25, which compares the predicted and simulated displacement responses at the fourth floor. A similarly strong match is observed for velocity responses in Figure 26. The evolution of the estimated stiffness is shown in Figure 27, where the prediction begins shortly after the damage instance. A slight delay is evident between the actual and predicted damage instances, which is attributed to the online SVD that requires a short accumulation window to reflect noticeable changes in the POM. This behavior is further supported by Figure 24, where the shape of the POM slope begins to evolve immediately after damage, but becomes distinctly identifiable approximately 3 s after the damage instance.

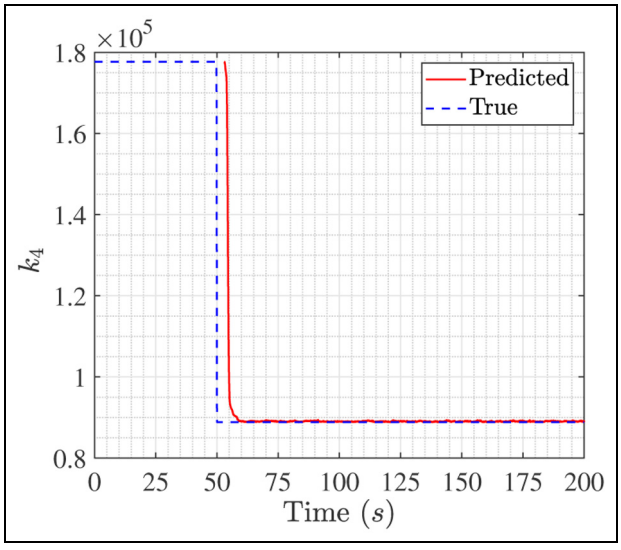


Figure 27. Comparison of true and predicted stiffness for a three-dimensional steel building.

Conclusions

This study presents a real-time framework for structural damage detection and state estimation by combining data-driven damage localization with Bayesian filtering, with direct relevance to DT-enabled SHM. Specifically, damage was identified by monitoring changes in the slope of the first POM, which was updated continuously using an online SVD algorithm. Simultaneously, UKF was employed to estimate the full structural state and update stiffness parameters only at the localized damage locations. This parallel operation of damage localization and parameter estimation enabled real time and efficient monitoring of the system. Compared to traditional methods that required full parameter estimation or relied on predefined damage models, the proposed approach offered several key advantages. First, localizing damage prior to estimation reduced computational complexity and enhanced numerical stability. Second, selective updating of parameters improved robustness in the prediction of parameters. Lastly, the real-time integration of online SVD and UKF supported early detection and continuous monitoring, making the framework well-suited for real-time SHM applications.

The effectiveness of the methodology was demonstrated through both numerical and calibrated three-dimensional steel building studies. The numerical investigation included an eight-story shear frame subjected to various damage scenarios, such as adjacent, non-adjacent, and progressive stiffness reductions. The calibrated study was carried out on a six-degree-of-freedom lumped mass model derived from a laboratory-scale three-dimensional steel building. To emulate realistic field conditions, white Gaussian noise was added to the displacement responses, with the noise variance set to 5% of the signal's variance. Despite the presence of measurement noise, the framework consistently achieves accurate damage localization and reliable estimation of the system states and stiffness parameters, demonstrating its practical relevance for continuous structural condition assessment.

A key limitation of the present framework is the assumption that the system is fully observable in terms of displacement responses. In real-world applications, such complete measurement coverage may be challenging due to restrictions on sensor deployment, instrumentation cost, and accessibility. Nevertheless, emerging non-contact measurement technologies particularly computer vision–based methods offer a promising pathway to obtain full displacement data with minimal physical instrumentation. In addition to this measurement limitation, the current implementation assumes that the input excitation is known, which may not always hold under ambient or operational loading. Future work will focus on relaxing both assumptions by integrating sparse-sensor strategies, computer vision-based displacement acquisition, and input estimation techniques. Further developments will also explore incorporating multiple POMs to enhance localization sensitivity and broaden the applicability of the framework to more complex structural configurations.

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Biswarup Bhattacharyya: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability statement

The underlying data supporting the findings of this study are available from the corresponding author upon a reasonable request.

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Appendix

Computational complexity analysis

It is essential to compare the computational complexity of the full SVD, the online SVD, the classical UKF (which estimates stiffness at all DOFs), and the proposed POD–UKF (which estimates only the parameters related to the damaged location). In dynamic systems where the response may change due to damage, the SVD must be recomputed at each time instant when new data arrive. If the response matrix grows as $\mathbf{R} \in \mathbb{R}^{N\times b}$ with $b = 1, 2, \ldots, n$, the cost of performing one full SVD at time b is approximately $\mathcal{O}(N^2 b + N^3)$ (for $N < b$). The cumulative cost of recomputing the full SVD at every time step becomes:

$$\begin{aligned} t_{\text{SVD}} &= N^2 \sum_{b=1}^{n} b + N^3 \\ &= N^2 \left(\frac{n(n+1)}{2} \right) + N^3 \end{aligned} \tag{41}$$

which grows quadratically with the number of snapshots and requires storing the entire dataset. In contrast, the online SVD updates the decomposition using only the newest snapshot and never revisits previous data. Its computational cost consists of the SVD update, projection operations, and occasional orthogonalization[72]:

$$\begin{aligned} t_{\text{SVD update}} &= 2N \left[\frac{n_a}{6} (n_a + 1)(2n_a + 1) - 1 + (n_a + 1)^2 (n - n_a) \right] \\ t_{\text{projection}} &= 2N(n_a + 2(n_a - 1)) \\ &\quad + 4N(n_a + 1)(n - n_a) - N(n - 1) \\ t_{\text{ortho}} &= N\, n_a^2. \end{aligned} \tag{42}$$

where $n_a \ll N$ is the number of retained POMs. These terms scale linearly with n and depend only on n_a, making the method highly efficient for real-time POM updates.

For the UKF, the augmented state has dimension $n_x = 2N + n_\theta$. The most expensive operation in each UKF step is the computation of the Cholesky factor of $P \in \mathbb{R}^{n_x \times n_x}$, and the number of arithmetic operations required for this factorization increases proportionally to $(2N + n_\theta)^3$. Generating the $2n_x + 1$ sigma points requires forming vectors of length n_x, so the effort increases proportionally to $(2N + n_\theta)^2$. Each sigma point is then propagated through the process model. Evaluating the matrix exponential $\exp(A_c\, \mathrm{d}t)$ requires work proportional to N^3. Since this exponential must be evaluated for every sigma point, the total propagation cost grows proportionally to $(2N + n_\theta)\, N^3$. The predicted state mean is obtained through a weighted

summation of all sigma points, and its computational effort increases proportionally to $(2N+n_\theta)^2$. The predicted covariance is constructed from outer products of vectors of length n_x, and therefore its computational effort grows proportionally to $(2N+n_\theta)^3$. The measurement model simply extracts the displacement components, resulting in a cost that increases proportionally to $N(2N+n_\theta)$. Forming the P_{yy} requires work proportional to $N^2(2N+n_\theta)$, and the inversion of P_{yy} requires work proportional to N^3. P_{xy} requires an amount of computation that grows with $N(2N+n_\theta)^2$, and the Kalman gain update grows proportionally to $N^2(2N+n_\theta)$. The dominant contributors to the total computational effort are the Cholesky factorization of P, the construction of the predicted covariance, and the repeated evaluation of the matrix exponential during sigma-point propagation. Each of these steps contains cubic dependence on the augmented state dimension, and hence the total computational effort of the UKF increases approximately with $(2N+n_\theta)^3$. Consequently, increasing the number of unknown parameters n_θ causes the computational cost per time step to rise approximately with n_θ^3. With the proposed POD–UKF approach, the computational burden can be significantly reduced by exploiting the fact that only the stiffness parameters associated with the damage location need to be included in the state vector. In a full N-DOF structural system, a classical UKF requires estimating all N stiffness parameters, which directly increases the state dimension and results in a per–time–step computational cost that grows approximately with N^3 due to the covariance and innovation matrix operations. In contrast, the POD–UKF formulation restricts the augmented state to include only the relevant stiffness parameters, denoted by n_θ, where $n_\theta \ll N$. As a result, the dominant UKF operations scale with n_θ^3 instead of N^3, providing a substantial reduction in computational cost while still accurately capturing the evolution of the damaged stiffness components.